

VALKENBURG, Netherlands

WMO: 062100

Lat: 52.1397N

Lon: 4.4364E

Elev: 1

StdP: 101.31

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-4.1	-9.7	1.6	-5.0	-7.2	2.1	-3.3	15.4	8.7	14.3	8.9	3.5	70	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	27.2	19.8	24.9	18.8	22.9	17.9	20.7	25.4	19.6	23.5	18.7	22.0	4.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.8	22.5	18.1	13.0	21.4	17.3	12.4	20.4	59.4	25.6	55.9	23.5	53.0	22.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	11.5	10.1	DB	-8.6	31.3	3.3	1.5	-11.0	32.3	-12.9	33.2	-14.7	34.0	-17.1	35.1	
			WB	-9.0	22.8	3.2	0.9	-11.3	23.4	-13.2	23.9	-15.0	24.4	-17.4	25.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	3.9	4.3	6.2	9.3	12.6	15.2	17.6	17.7	15.0	11.5	7.8	4.5
	DBStd	5.91	4.07	3.69	3.08	3.38	3.16	2.76	2.68	2.43	2.34	3.01	3.27	4.14
	HDD10.0	805	189	161	123	53	9	0	0	0	0	19	78	173
	HDD18.3	2915	447	393	377	272	180	101	47	42	103	211	314	428
	CDD10.0	991	1	1	4	32	90	157	236	238	151	66	13	3
	CDD18.3	58	0	0	0	0	2	7	24	22	3	0	0	0
	CDH23.3	367	0	0	0	2	24	57	148	121	13	1	0	0
	CDH26.7	76	0	0	0	0	3	12	35	26	1	0	0	0
Wind	WSAvg	4.7	5.6	5.4	5.2	4.5	4.5	4.5	4.2	4.0	4.2	4.5	4.7	5.3
Precipitation	PrecAvg	928	78	68	53	43	57	68	85	105	91	91	101	89
	PrecMax	1192	170	131	120	104	114	150	195	234	250	161	191	160
	PrecMin	669	9	20	9	0	20	22	21	19	12	17	8	18
	PrecStd	127	38	30	29	26	27	37	45	60	53	40	43	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.3	17.0	23.0	26.1	28.5	30.1	29.4	25.1	20.6	16.1	13.5
		MCWB	10.7	9.8	12.1	14.7	17.5	20.1	20.9	20.7	19.4	17.0	14.2	11.9
	2%	DB	11.1	11.2	14.0	19.5	23.2	24.8	27.2	26.7	22.4	18.2	14.4	12.2
		MCWB	9.8	9.2	10.6	13.6	16.1	18.4	19.9	20.0	18.2	15.8	12.9	10.9
	5%	DB	9.9	10.1	12.1	16.9	20.6	22.1	24.7	24.0	20.5	16.9	13.4	11.1
		MCWB	8.7	8.5	9.6	12.2	15.1	17.3	19.2	19.0	17.0	15.0	12.1	10.0
	10%	DB	8.9	9.1	10.7	14.6	18.1	20.2	22.5	22.0	18.9	15.9	12.3	9.9
		MCWB	7.8	7.8	8.6	11.2	13.9	16.2	18.2	18.1	16.2	14.1	10.9	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.8	12.8	15.6	19.2	21.2	22.4	22.0	20.0	17.5	14.3	12.3
		MCDB	12.0	12.3	16.1	21.2	24.5	26.7	27.0	27.6	23.7	19.9	15.3	13.1
	2%	WB	10.0	9.6	11.1	14.0	16.9	19.1	20.7	20.5	18.6	16.2	13.2	11.1
		MCDB	11.0	10.8	13.3	18.4	21.0	23.2	25.5	24.9	21.7	17.8	14.3	12.0
	5%	WB	8.8	8.7	9.9	12.7	15.5	17.8	19.6	19.5	17.6	15.3	12.2	10.1
		MCDB	9.7	9.9	11.7	16.3	19.6	21.3	23.5	23.0	19.9	16.7	13.2	11.0
	10%	WB	7.9	7.9	8.9	11.5	14.3	16.6	18.6	18.6	16.7	14.4	11.1	8.9
		MCDB	8.8	9.0	10.3	14.1	17.7	19.5	21.9	21.5	18.6	15.7	12.1	9.9
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.9	6.0	7.5	7.5	7.3	6.9	7.2	6.8	6.0	4.8	4.3
		MCDBR	4.3	5.4	8.8	11.2	12.0	11.3	11.0	10.8	9.1	6.8	5.0	4.1
		MCWBR	4.0	4.1	6.0	6.4	6.4	6.2	5.4	5.4	5.5	5.0	4.4	4.0
	5% WB	MCDBR	4.1	4.7	7.6	10.1	10.6	9.8	9.6	9.3	8.1	6.5	4.8	4.2
		MCWBR	4.0	4.0	5.5	6.1	6.2	5.9	5.2	5.1	5.3	5.2	4.3	4.2
Clear-Sky Solar Irradiance	taub	0.342	0.362	0.395	0.418	0.424	0.419	0.428	0.419	0.397	0.377	0.354	0.337	
	taud	2.312	2.305	2.231	2.213	2.257	2.287	2.273	2.312	2.361	2.367	2.333	2.306	
	Ebn at Noon	639	739	785	815	829	836	821	807	779	714	616	572	
	Edh at Noon	65	86	111	127	128	125	125	115	98	80	63	56	
All-Sky Solar Radiation	RadAvg	0.70	1.39	2.61	4.22	5.12	5.47	5.18	4.35	3.15	1.80	0.81	0.53	
	RadStd	0.07	0.26	0.31	0.41	0.35	0.37	0.55	0.35	0.27	0.15	0.08	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page